Sahaj Bhadja

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Summary

Customer-focused Application Engineer with 2+ years delivering **technical demonstrations, workflow implementation, training support, troubleshooting, and data analysis** across configurable software products, integrations, and cross-functional teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Translated stakeholder workflows into FastAPI services and dashboards, processing 1K+ concurrent AI jobs at 3x throughput through **technical discovery, solution implementation, monitoring, and end-to-end customer-focused application ownership**.
- Optimized indexed Azure SQL queries and row-level isolation for 500+ users, cutting latency from 1.2 seconds to 350 milliseconds through **workflow analysis, application optimization, data validation, and production troubleshooting**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **user feedback, demonstrations, REST integration, and iterative releases**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **systematic troubleshooting, testing, and workflow validation**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform services and data integrations using object-oriented patterns, AWS Glue, S3, and caching, improving response times 30% while reducing technical debt 20% across customer-facing application workflows.
- Automated 20+ unit and integration suites across Windows, Linux, and macOS using PyTest, Mockito, Jenkins, and virtual machines, strengthening **issue reproduction, release validation, documentation, and cross-team technical support**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React and Node.js services for resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, configurable workflows, reusable interfaces, and role-aware product experiences** across locations.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while using Sentry, Azure Monitor, and Kibana for **application diagnostics, incident resolution, workflow reliability, and successful customer releases**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Led requirement discovery and presented solution architecture for 12+ integrations, mapping booking, payment, coupon, and status workflows to **explain technical concepts, collect feedback, and reduce design errors 25%**.
- Built Flask APIs supporting 50K+ records with sub-200-millisecond responses while providing **product demonstrations, implementation guidance, technical status updates, and customer support** for booking and payment workflows during reviews.

Skills

Application Support: Technical Discovery, Product Demonstrations, Customer Training, Presentations, Troubleshooting, Implementation, Workflow Optimization
Laboratory Technology: Laboratory Informatics, LIMS, Instrument Data Workflows, Sample Tracking, Data Analysis, Reporting, Workflow Validation
Software & Data: Python, JavaScript, TypeScript, React, Flask, Node.js, REST APIs, JSON, SQL, PostgreSQL, MySQL, Azure SQL
Cloud & Operations: AWS, Azure, Docker, Jenkins, CI/CD, Git, Sentry, Kibana, Azure Monitor, Linux, Incident Management
Collaboration: Requirements Gathering, Technical Documentation, Cross-Functional Collaboration, Customer Feedback, User Adoption, Problem Solving, Communication, Agile

Projects

LabFlow Support Console | *Python, React, PostgreSQL, REST APIs, Docker*

- Designed a **laboratory workflow support application** for sample intake, instrument readiness, method assignment, exception tracking, results review, and audit documentation, with guided troubleshooting and role-specific training views.

Instrument Insight Studio | *Python, Pandas, React, SQL, Plotly*

- Built an **instrument performance demonstration and analytics toolkit** that validates run data, visualizes utilization and quality trends, flags workflow anomalies, and generates presentation-ready reports for consultative customer reviews.

Chat2SQL | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker* | Code

- Built an **AI-powered SQL development workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript interface with Docker-based deployment.